ARM PROCESSOR AND ITS APPLICATIONS

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PROBLEM STATMENT

SMART NIGHT LIGHT

- **Introduction :** Lighting systems play a crucial role in ensuring safety, security, and comfort in residential, commercial, and public spaces. However, conventional night lighting systems often contribute to excessive energy consumption and carbon emissions due to their continuous operation and dependence on grid electricity. In regions with limited or unreliable power infrastructure, maintaining consistent nighttime illumination remains a significant challenge.

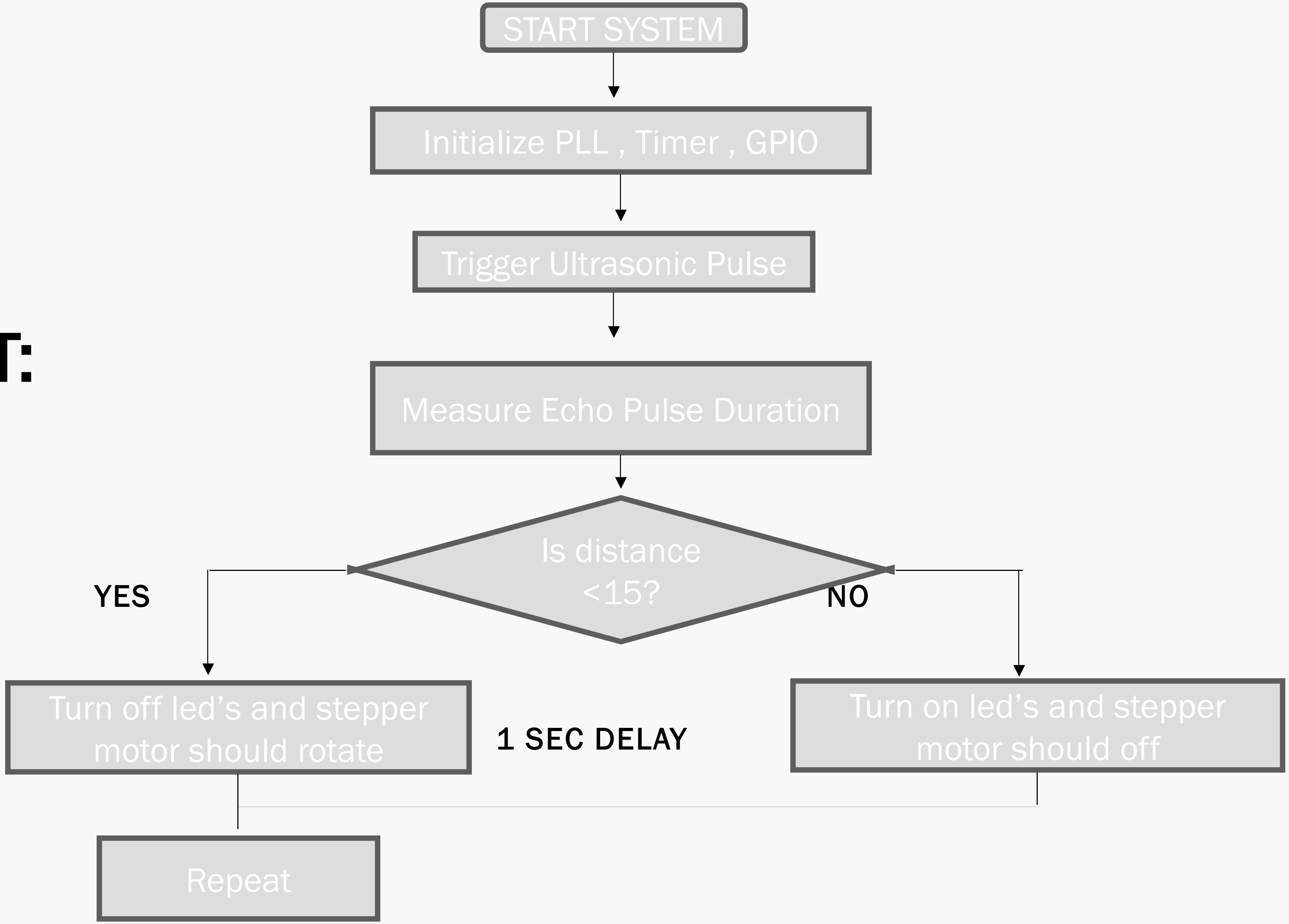
PROBLEM ANALYSIS:

Design and implement a smart night light system using an LPC2148 microcontroller that :
Detects the presence of an object within a specified range using an ultrasonic sensor .
Responds by turning ON LEDs and rotating a stepper motor clockwise when an object is detected nearby .
Otherwise, the system keeps the LEDs OFF and rotates the stepper motor anticlockwise

Existing Scenario / Traditional System :
Regular night lights use LDR (Light Dependent Resistor) to detect darkness . They do not respond to human/object motion or distance .
They lack interactive feedback (no movement or directional light).

Proposed Solution : Replace LDR with an ultrasonic distance sensor for more intelligent and accurate detection . Add a stepper motor to perform physical movement (e.g., turn a light or camera).

FLOW CHAT:


START SYSTEM
Initialize PLL , Timer , GPIO
Trigger Ultrasonic Pulse
Measure Echo Pulse Duration
Is distance <15?
YES
NO
Turn off led’s and stepper motor should rotate
1 SEC DELAY
Turn on led’s and stepper motor should off
Repeat

CODE:

```
#include <lpc214x.h>
#define TRIG (1 << 5)              // P0.5
#define ECHO (1 << 4)              // P0.4
#define STEPPER_MASK (0xF << 12)  // P0.12 - P0.15 stepper
motor pins
#define LED (1 << 16)              // P0.16 for LED
#define PLOCK 0x00000400
void initTimer0(void) {
    T0CTCR = 0x0;
    T0PR = 59;       // 1us tick (for 60 MHz PCLK)
    T0TCR = 0x02;    // Reset timer
}

void delayUS(unsigned long us) {
    T0TCR = 0x02;    // Reset
    T0TCR = 0x01;    // Enable
    while (T0TC < us);
    T0TCR = 0x00;    // Disable
}
```

```
VOID DELAYMS(UNSIGNED INT MS) {
    DELAYUS(MS * 1000);
}
VOID STARTTIMER0(VOID) {
    T0TCR = 0X02;
    T0TCR = 0X01;
}
UNSIGNED INT STOPTIMER0(VOID) {
    T0TCR = 0X00;
    RETURN T0TC;
}
VOID SETUPPLL0(VOID) {
    PLL0CON = 0X01;
    PLL0CFG = 0X24;
}
VOID FEEDSEQ(VOID) {
    PLL0FEED = 0XAA;
    PLL0FEED = 0X55;
}
VOID CONNECTPLL0(VOID) {
    WHILE (!(PLL0STAT & PLOCK));
    PLL0CON = 0X03;
}
VOID INITCLOCKS(VOID) {
    SETUPPLL0();
    FEEDSEQ();
    CONNECTPLL0();
    FEEDSEQ();
    VPBDIV = 0X01;
}
```

```
void rotateStepper(void) {
    unsigned int i;
    for (i = 0; i < 50; i++) {
        IO0CLR = STEPPER_MASK;    // Clear all stepper pins
        IO0SET = (1 << 12);       // Step 1
        delayMS(10);
        IO0CLR = STEPPER_MASK;
        IO0SET = (1 << 13);       // Step 2
        delayMS(10);
        IO0CLR = STEPPER_MASK;
        IO0SET = (1 << 14);       // Step 3
        delayMS(10);
        IO0CLR = STEPPER_MASK;
        IO0SET = (1 << 15);       // Step 4
        delayMS(10);
    }
    IO0CLR = STEPPER_MASK;        // Turn off all stepper pins at end
}
```

```
int main(void) {
    int echoTime = 0;
    float distance = 0;
    initClocks();
    initTimer0();
    IO0DIR |= TRIG;             // P0.5 as output
    IO0DIR &= ~ECHO;            // P0.4 as input
    IO0DIR |= STEPPER_MASK;     // P0.12 to P0.15 as outputs
    IO0DIR |= LED;              // P0.16 as output for LED
    IO0CLR = TRIG | STEPPER_MASK | LED; // Clear all pins initially
    delayMS(2000);              // Initial delay
    while (1) {
        IO0SET = TRIG;
        delayUS(10);
        IO0CLR = TRIG;
        while (!(IO0PIN & ECHO)); // Wait for echo to go high
        startTimer0();
```

```
while (IO0PIN & ECHO);    // Wait for echo to go low
        echoTime = stopTimer0();
        distance = (0.0343 * echoTime) / 2;
        if (distance > 0 && distance < 15.0) {
            IO0SET = LED;            // Turn ON LED
            rotateStepper();         // Rotate stepper motor
        } else {
            IO0CLR = LED;            // Turn OFF LED
            IO0CLR = STEPPER_MASK;   // Stop stepper motor
        }

        delayMS(1000); // Wait 1 second before next reading
    }
}
```

SPECIFICATIONS OF ALL PERIPHERALS USED:

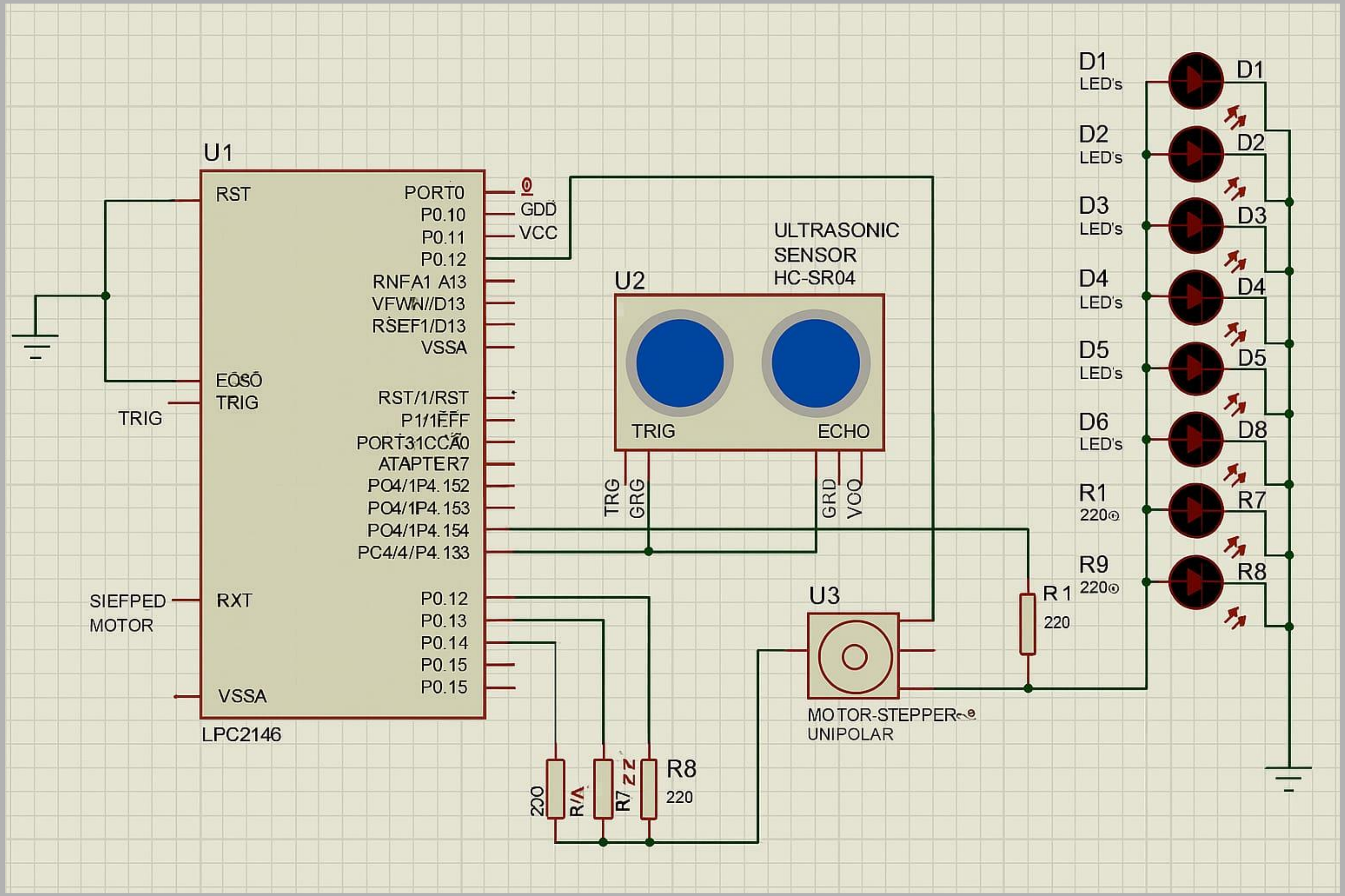
Hardware Components

Microcontroller	LPC2148 ARM7TDMI-based controllerller
Ultrasonic Sensor	HC-SR04 (or compatible) for distance measurement
Stepper Motor	Unipolar stepper motor (connected to P0.12– P0.15)
LED	1 LED connected to P0.16 (t odication)
Power Supply	5V regulated supply for logic and senssors/motors
Timer Used	Timer0 for microsecond timing during ultrasonic echo mea-

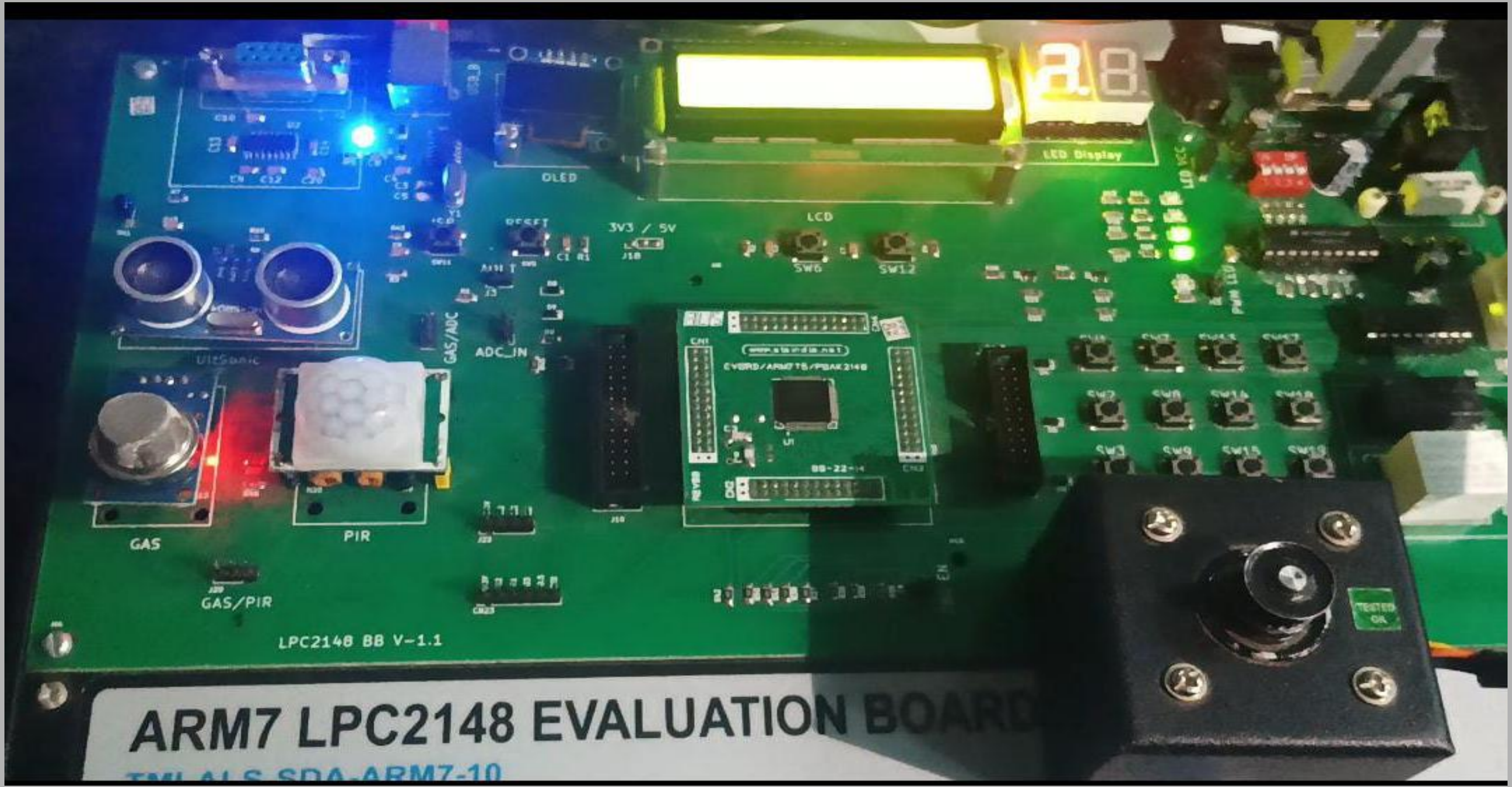
Software / Functional Specifications

Object Detection	Detects object within 15 cm using ultrastnc sensor
LED Control	LED OFF when object is near, ON otherwise
Motor Direction	Always rotate stepper motor clockwise when object is near
Distance Calculation	distance = (echoTime × 0.0343) / 2
Timer Resolution	1 us using Timer0 wih TOPR × 9 (assuming 60 MHz PCLK)
Delay Control	Software delays using Timer0 (delayUS() and delayMS))
Rotation Count	50 steps per detection loop (4-step sequence × 50 iterations = 200 total steps)

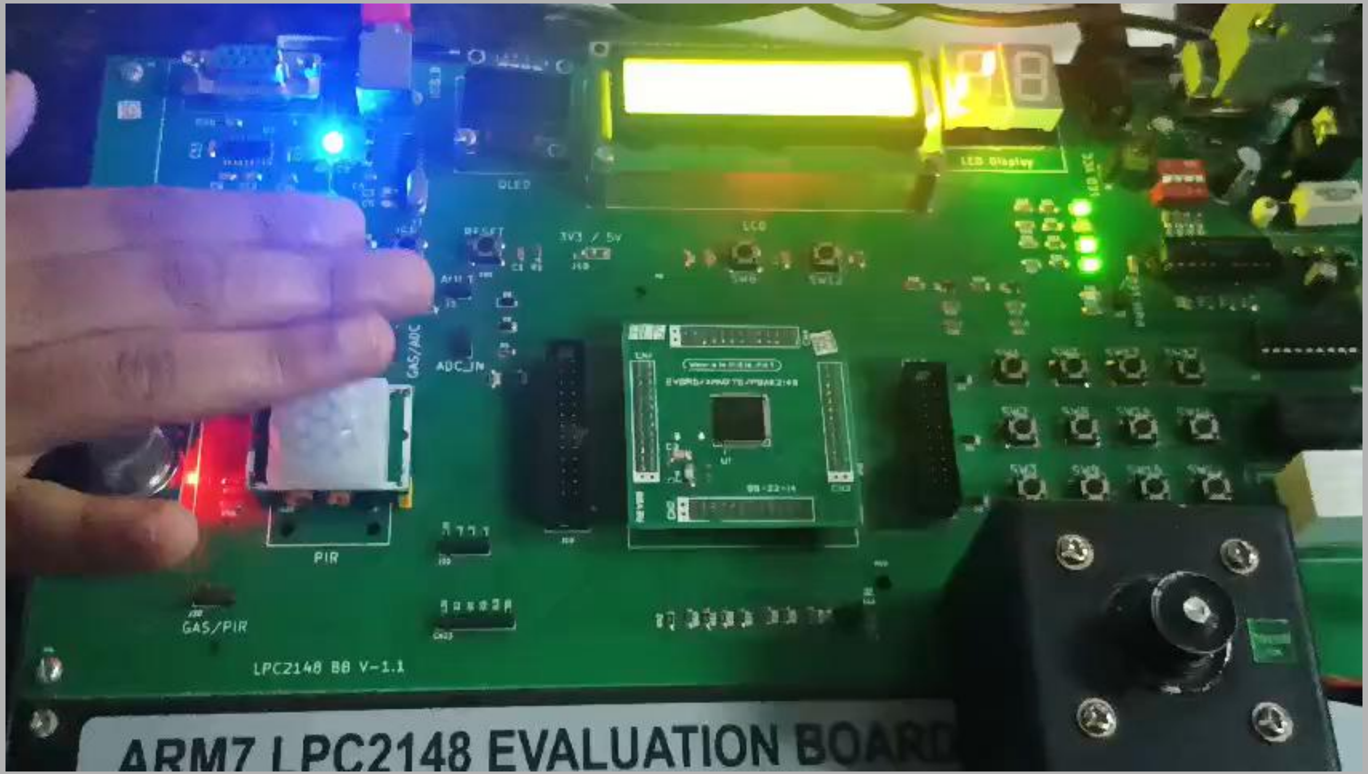
SIMULATION:


U1
RST
PORT0
P0.10
P0.11
P0.12
VSSA
TRIG
RXT
P0.12
P0.13
P0.14
P0.15
P0.15
VSSA
LPC2146
SIEFPED
MOTOR
TRIG
GDD
VCC
ULTRASONIC
SENSOR
HC-SR04
U2
TRIG
ECHO
U3
MOTOR-STEPPER
UNIPOLAR
R8
220
R7
R1
220
D1
LED's
D2
LED's
D3
LED's
D4
LED's
D5
LED's
D6
LED's
R1
R9
D1
D2
D3
D4
D5
D8
R7
R8

RESULTS:


ARM7 LPC2148 EVALUATION BOARD
LPC2148 BB V-1.1
GAS
PIR
GAS/PIR
OLED
LCD
LED Display
RESET
3V3 / 5V
ADC_IN
GAS/ADC
SW6
SW12

RESULTS:


OLED
RESET
ADC_IN
PIR
GAS/PIR
LPC2148 BB V-1.1
ARM7 LPC2148 EVALUATION BOARD

RESULTS:

The implemented embedded system successfully integrates an ultrasonic sensor, stepper motor, and LEDs using the LPC2148 microcontroller. When an object is detected within a 15 cm range, the system triggers the clockwise rotation of the stepper motor and turns ON the LEDs, providing a clear visual and mechanical response. If no object is detected, the motor rotates anticlockwise and LEDs remain OFF.

THANK YOU